

Optimal Segments Selection for EEG Classification

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Abstract— Electroencephalography is non invasive technique used to measure electrical brain activity and to collect EEG signals, through surface electrodes properly positioned on the scalp, which analyzed and interpreted in BCI (Brain Computer Interface) allow the comprehension of human intentions of movement. This is particularly useful for persons with severe physical handicaps. In this paper dataset of motor imagery is used to describe left and right hand movement imagery. To extract discriminating features from EEG signals, wavelet transform is a concrete method which deals with time frequency aspect of non stationary EEG signals. Indeed, the creation of statistical parameters describing wavelet coefficients to build features vectors gives good classification results when used with Linear Discriminate Analysis (LDA) or Neural Networks.

Keywords—component; EEG, Wavelet Transform, Linear Discriminate Analysis, and Neural Networks.

I. INTRODUCTION

EEG signals, used in brain computer interface (BCI), represent a prospect for communication which enables severely handicapped persons to have a new way to translate their intentions of movement and mental activities and to control external devices with brain activity; actuality classification of these signals allows the discrimination of various movements or even intentions of movement realized by a person [16], [8]. Thus, to achieve EEG classification in BCI, generally in the beginning a pre-treatment of EEG signals is done, then the extraction features describing EEG signals is made, and eventually the movement classification is achieved.

The choice of adequate method for features extraction is critical to obtain better classification results, several techniques are used for features extraction in BCI and described in literature [9] ; band power [17], common spatial pattern (CSP) [11] or autoregressive models (AR) [4], [13], [14], but these methods detect merely information of time, frequency or spatial domain. An interesting approach which covers the time frequency properties of EEG signals is wavelet transform [2]. Indeed, in this paper this method is used to capture information from two channels of EEG signals located over Centro-temporal region of the scalp. Actually, temporal filtering of EEG trials is performed to extract features from EEG segments and find segments having the highest classification accuracies. Then statistical parameters extracted from wavelet coefficients are fitted as input to the classifiers. Two classifiers are used in this case; Linear Discriminate Analysis (LDA) and a Neural

Networks, to compare the performance of classification of left and right hand movement.

II. EEG DATASET DESCRIPTION:

The purpose of this analysis is to differentiate two motor imagery tasks; the movement of left and right hand labeled respectively as class 1 and class 2; to determine temporal segments reaching highest classification results. Signals recorded from EEG channels C3 and C4, according to the 10 – 20 standard system of electrode placement, over sensory-motor cortices; activated during movement or intension to make mental or physical tasks; are analyzed in order to assess sensory-motor μ rhythms [8 13] HZ [13] and Beta [13 30] HZ located in the Centro-temporal region. Indeed, the preparation or the imagination of a movement is accompanied by a decrease of power, named Event-Related Synchronization (ERD), in certain frequency bands specifically in Mu and Beta rhythms. After movement there is a rise of power in these frequency bands indicating a relaxation due to the Event-Related De-synchronization (ERS).

EEG motor imagery data set utilized in this study is dataset III publicly available in BCI competition II (2003) and provided by Department of Medical Informatics, Institute for Biomedical Engineering, University of Technology Graz; this data is described on [12]. This data contains a train dataset and a test dataset composed of 140 trials of 9s each. From $t=3s$ to 9s the subject was asked to perform left or right hand movement imagery. EEG signals are sampled with 128Hz and filtered between 0.5 and 30Hz.

x_train data contains 140 trials of 9s each of training data, including 3 EEG signals measured over channels C3, CZ and C4. y_train data is composed of the labels of trials saved in x_train which are {1,2} respectively for left and right hand movement classes.

III. EEG TEMPORAL WINDOWING:

In each EEG trial, temporal segments of [3s-8s] related to movement tasks are considered and are windowed to 256 samples windows with a shift of 1 point. This implies that total number of segments in each trial presented for features extraction is $N=385$. This temporal windowing is applied to 140 trials in the train set and 140 other trials in the test set. Segments are grouped according to their numbers in each EEG trial, so that each subset of segments have the same number in

all trials, example subset1 is composed by segments number 1 assembled from all 140 trials.

IV. FEATURES EXTRACTION METHOD:

To create discerning features in the frequency-domain, EEG analysis is complex and time-frequency techniques are required [1]. Therefore Discrete Wavelet Transform DWT is applied to characterize EEG signals recorded from channels C3 and C4.

A. Discrete Wavelet Transform:

In this paper, discrete wavelet transform is employed to decompose EEG segments into frequency sub-bands using different mother wavelet functions.

Tests are performed with different types of wavelet functions; Daubechies [2], Symmlet and Coiflet; in order to find the one giving maximum efficiency and to investigate the influence of computed coefficients on the classification rate. In order to improve the time-frequency resolution and to make it satisfactory by selecting suitable wavelet function, in each of the mentioned wavelet families above [15], different wavelet functions are used as following:

1. Daubechies4, Daubechies6, Daubechies8, Daubechies10, and Daubechies12.
2. Coiflet2, Coiflet4, and Coiflet5.
3. Symmlet5, Symmlet6, Symmlet7, Symmlet8, Symmlet9, and Symmlet10.

Wavelet transform is applied to extract quantitative time-frequency parameters from EEG signals [3], [6]. Wavelet coefficients are calculated for each EEG segment. The computed wavelet coefficients show the energy distribution in time and frequency of EEG signals. Thus, EEG signals with a sampling frequency of 128 Hz are decomposed into three levels; details D1–D3 and one final approximation A3 as shown in Table 1.

Subsequently parts of the signal that correlates well with the frequencies of interest [8 32] HZ covering μ and beta rhythms, necessary for classification, are retained in the wavelet coefficients of details D2 and D3. Then, only following statistical parameters are considered in order to reduce the number of extracted features [7]:

1. Mean of the absolute values of wavelet coefficients in each sub-band.
2. Standard deviation of coefficients in each sub-band.
3. Average power of the wavelet coefficients in each sub-band.

Parameters 1 and 3 provide information related to the power spectrum of the signal, and parameter 2 presents the changes in the power spectrum over time. Statistical parameters obtained from coefficients of details sub-bands D2 and D3 are concatenated to form six features related to EEG signals

TABLE I. FREQUENCIES CORRESPONDING TO DIFFERENT LEVELS OF DECOMPOSITION

Decomposed signal	Frequency range(HZ)
D1	32- 64
D2	16- 32
D3	8- 16
A3	0- 8

recorded from channels C3 and C4. Ultimately, $6*2=12$ features vector based on statistical parameters of wavelet coefficients are grouped into vectors and normalized to characterize every EEG segment.

EEG trials are windowed using windows of 256 samples with a shift of 1 point. Hence, each EEG trial is composed of a total number of 385 segments. Every EEG segment is characterized by a vector of 12 normalized features. Segments are grouped into subsets according to their number in each trial, for example all segments having number 1 and obtained from trials numbers 1 to 140 are grouped into subset number 1. So, to generalize segments having number k and obtained from all 140 trials are grouped into a subset number k, where k is an integer from 1 to N, and N=385 is the total segments number in one EEG trial. Then, the creation of 385 matrices to represent the features extracted from all EEG segments subsets is achieved. In this manner, every features matrix is characterizing an EEG segments subset, and is composed of 12 rows representing features characterizing each EEG segment included in this subset and 140 columns indicating 140 trials.

Feature extraction is obviously performed the same way to the train and test datasets. Accordingly, each matrix related to a subset of EEG segments is presented as input to the chosen classifiers.

V. CLASSIFICATION METHOD:

To release EEG classification, two classifiers are utilized; Linear Discriminate Analysis LDA and Neural Networks. Thus, classification results are compared for different feature extraction methods.

A. Linear Discriminate Analysis LDA:

LDA is a linear classifier suitable for EEG classification due to the advantages of simplicity, stability and insensitivity to over fitting. In the proposed system, each segments subset obtained one features matrix which is then used to classify the corresponding state {0, 1}; referring respectively to left and right hand movement; using linear discriminate analysis. LDA using mahalanobis distance is applied to classify the two given classes based on the assumption that both classes are under normal distribution. LDA performance is measured by the classification accuracy as following:

Accuracy= [number of correctly classified patterns / total number of tested patterns]*100.

Fig. 1 showed moving average classification accuracy evolution in different EEG segments using LDA classifier and features extracted using different wavelet functions including DB10, Coiflet5 and symmlet10.

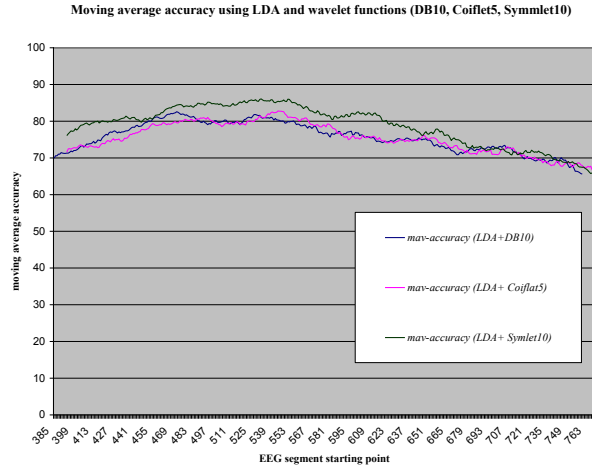


Figure 1. Moving Average Accuracy of LDA and features extracted by wavelets functions (DB10, Coiflet5, and Symmlet10).

B. Neural Networks:

In this paper, Multilayer feed forward neural network MLP [10] is used, in this network the information progress in forward direction from the input nodes, through the hidden nodes and to the output nodes [5]. Multilayer perceptron MLP is a simple type of neural networks including at least one hidden layer. The architecture of the chosen neural network consist on creating an input layer of 12 nets representing EEG features extracted from an EEG segment, a single hidden layer of 20 nets, and one output layer of one net as shown in Fig. 3. Output layer exploit linear transfer function. Multi-layer neural network MLP is employing Backpropagation training algorithm Levenberg-Marquardt for classification allowing rapid execution of the trained network. Indeed, the mean square error MSE between network output and desired outputs is evaluated and nets weights are adjusted to minimise MSE. Training was conducted until the mean square error MSE (between desired and target) fell below 0.01 or reached a maximum iteration limit of 1000. The mean square error refers to the error limit to stop NN training. The desired target output was set to {1, 0} for left hand movement or class 1 and right hand movement or class 2. Patterns from 140 trials are used in training session and 140 trials are reserved for test. Each trial is presented by features extracted from a total of 385 segments. The network performance is estimated in terms of classification accuracy based on the percentage of correct recognition rate of hand movement classes in EEG segments subsets, calculated as following:

Accuracy= (number of correctly classified patterns / total number of tested patterns)*100.

Fig. 2 showed evolution of the moving average classification accuracy in different EEG segments using MLP classifier and features extracted via different wavelet functions including DB10, Coiflet5, and Symmlet6.

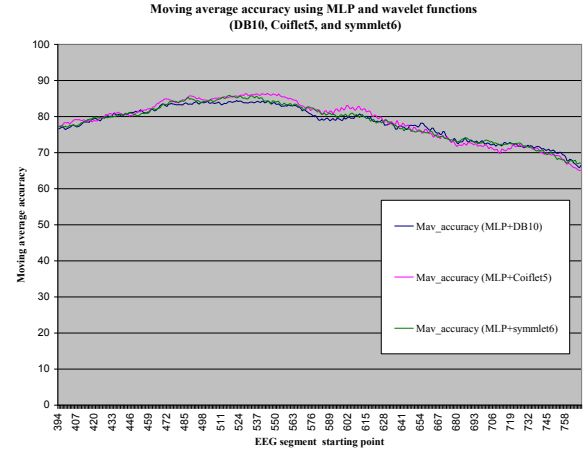


Figure 2. Moving Average Accuracy of MLP and features extracted by wavelet functions (DB10, Coiflet5, Symmlet6).

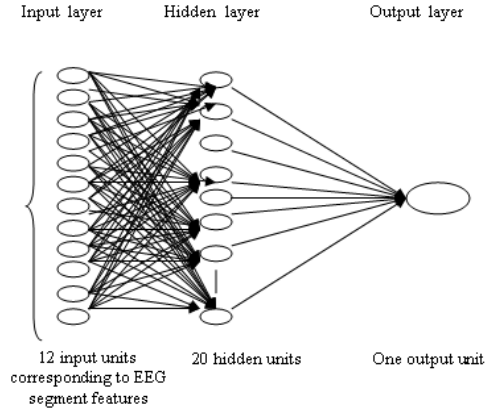


Figure 3. Architecture of neural network classifier

VI. RESULTS AND DISCUSSION:

In this paper, EEG segments were processed by different wavelet functions in order to select as far as possible essential features of motor imagery.

Fig. 1 and Fig. 2 described moving average accuracy evolution every 10 passing segments of features extracted by Daubechies, Coiflet, and Symmlet wavelet families and classified respectively by LDA and MLP classifiers as illustrated in (1):

$$Mav - Accuracy(k) = 1/10 * \sum_{i=k-9}^{i=k} (Accuracy(i)) \quad (1)$$

Results in Table 2 showed that maximum moving average accuracy reached 84.43% in segment (523 samples – 779 samples) using MLP classifier and Daubechies Wavelet function of order 10 (DB10). Therefore, the moving average accuracy; accomplished by features extracted via DB10 wavelet function and MLP classifier; was higher than 80% in segment (430 – 836 samples) as shown in Fig. 2.

TABLE II. HIGHEST CLASSIFICATION RESULTS ACHIEVED USING MLP CLASSIFIER FOR DIFFERENT WAVELET FAMILIES.

Wavelet family	Wavelet function	Optimal segments (samples)	Maximum moving average accuracy
Daubechies	DB10	523-779	84.43%
Coiflet	Coiflet5	553- 809	86.29%
Symmlet	Symmlet6	518- 774	85.86%

Results in Table 4 showed that LDA classifier achieved a maximum moving average accuracy of 82% in segment (476-732 samples) using Daubechies Wavelet function of order 10 (DB10). Then, moving average accuracy; accomplished by features extracted via DB10 wavelet function and LDA classifier; is higher than 80% in segment (465 samples – 754 samples) according to Fig. 1.

Considering Table 3 and Table 5, means of moving average accuracies using different Daubechies wavelet functions; DB4, DB6, DB8, DB10, and DB12; is calculated for both MLP and LDA classifiers, and attained respectively 84.37% and 81.44%. Thus, MLP classifier accomplished better classification results than LDA for the different Daubechies wavelet functions. Daubechies wavelet function of order 10 gives the highest classification results for both LDA and MLP classifiers. Classification accuracies reach **89%** using as input features extracted via DB10 wavelet function from segment (560-816 samples) and MLP classifier.

Take into consideration results in Table 2, moving average classification accuracy of 86.29% was accomplished using MLP and features extracted from segment (553-809 samples) via Coiflet5 wavelet. Fig. 2 showed that moving average accuracy of Coiflet5 extracted features and MLP classifier is higher than 80% in segment (428 – 834 samples).

Results in Table 4 showed that LDA classifier achieved a maximum moving average accuracy of 83.64% in segment (543- 799 samples) using Coiflet5 wavelet function. Fig. 1 showed that moving average accuracy values accomplished by features extracted via Coiflet5 wavelet function and LDA classifier; are higher than 80% in segment (477– 823 samples).

Observing Table 3 and Table 5, the mean of moving average accuracies for Coiflet2, Coiflet4, and Coiflet5, reached respectively 85.64% and 82.5% using MLP and LDA classifiers. Consequently, MLP classifier achieved higher classification results than LDA for different Coiflet wavelet functions (Coiflet2, Coiflet4, and Coiflet5). Coiflet5 wavelet function attained the highest classification results for both LDA and MLP classifiers. Classification accuracy reached 89% using as input features extracted via Coiflet5 wavelet function from segment (545- 801 samples) and MLP classifier.

TABLE III. MEAN OF MOVING AVERAGE CLASSIFICATION ACCURACIES USING MLP CLASSIFIER AND DIFFERENT WAVELET FAMILIES (DAUBECHIES, COIFLET, AND SYMMLET).

Wavelet family	Mean of moving average accuracy
Daubechies	84.37%
Coiflet	85.64%
Symmlet	84.88%

TABLE IV. HIGHEST CLASSIFICATION RESULTS ACHIEVED USING LDA CLASSIFIER FOR DIFFERENT WAVELET FAMILIES.

Wavelet family	Wavelet function	Optimal segments (samples)	Maximum moving average accuracy
Daubechies	DB10	476- 732	82%
Coiflet	Coiflet5	543- 799	83.64%
Symmlet	Symmlet10	552- 808	86%

Table 2 illustrated the optimal EEG segment achieving highest moving average classification accuracy using Symmlet wavelet functions for features extraction and MLP classifier. Highest classification result was 85.86% obtained for Symmlet6 in EEG segment (518- 774 samples). Taking into account Fig. 2, moving average classification accuracy is higher than 80% in segment (429- 835 samples).

Results in Table 4 showed that LDA classifier achieved a maximum moving average accuracy of 86% in segment (552-808 samples) using Symmlet10 wavelet function. Fig. 1 showed that moving average accuracy values accomplished by features extracted via Symmlet10 wavelet function and LDA classifier; are higher than 80% in segment (428- 834 samples).

Observing Table 3 and Table 5, the mean of moving average accuracy for Symmlet5, Symmlet6, Symmlet7, Symmlet8, Symmlet9, and Symmlet10 reached 84.88% and 82.35% using respectively MLP and LDA classifiers.

Consequently, LDA classifier achieved higher classification results than MLP for different Symmlet wavelet functions (Symmlet5, Symmlet6, Symmlet7, Symmlet8, Symmlet9, and Symmlet10). Classification accuracy reach 88% using as input features extracted via Symmlet10 wavelet function from segment (483- 739) samples and LDA classifier.

Classification results using MLP showed that moving average accuracy is higher than 80% for segments (430-836samples), (428- 834 samples) and (429- 835 samples) for features extracted respectively using DB10, Coiflet5 and Symmlet6 wavelet functions. In this manner, best classification results are obtained in EEG segment (429(±1) - 835(±1) samples) for MLP classifier. For LDA classifier highest classification results are also obtained using features extracted by Symmlet10 wavelet function from EEG segment (428-834 samples). Consequently features extracted from this segment (429(±1) - 835(±1) samples) can achieve good classification rates using appropriate wavelet functions for features extraction.

TABLE V. MEAN OF MOVING AVERAGE CLASSIFICATION ACCURACIES USING LDA CLASSIFIER AND DIFFERENT WAVELET FAMILIES (DAUBECHIES, COIFLET, AND SYMMLET).

Wavelet family	Mean of moving average accuracy
Daubechies	81.44%
Coiflet	82.5%
Symmlet	82.35%

VII. CONCLUSION

In this paper, EEG trials are windowed and prepared for features extraction by temporal windowing of 256 samples with a shift of one sample. The purpose is to find EEG segments giving highest classification accuracy in all trials. Feature extraction techniques based on different wavelet types are applied in order to compare their effects on the classification accuracy, among them; Daubechies10 and Coiflet5 wavelet functions reached good classification rates using both MLP and LDA classifiers. Symmet10 achieved higher results then both of them using LDA classifier, and attaining 86% of moving average classification accuracy. Considering the mean of moving average accuracy achieved by Daubechies, Coiflet and Symmlet wavelet families, MLP classifier offered slightly better results than LDA classifier. Classification accuracies of EEG segments are higher than 80% in segment (429($\pm$ 1) - 835($\pm$ 1) samples). Hence, extracted features from this temporal segment provide good classification results, and extracted patterns are efficient to discriminate classification classes. Obviously, in this case, subject capacity of movement imagination improved along time, and its concentration is the highest in this segment. Then a decrease in the classification rates is observed. So, tests achieved to find the most appropriate wavelet function for features extraction, and the fluctuation of classification accuracies in time in different EEG segments indicated that classification results depended on individual subject concentration variation over time. Discrimination of hand movement is achieved through LDA and MLP classifiers, results proved that efficiency of both of them is almost similar using earlier extracted features.

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